

CHAPTER 7

Environmental Conditions and How to Adapt

Difficulty is part of ultrarunning's appeal. The length of the event, the unknown time it will take to get from aid station to aid station, and running over varied terrain at night all present challenges to even the most experienced ultrarunners. However, two conditions in particular strike fear into ultrarunner's hearts: heat and altitude.

Heat and altitude affect runners similarly: they strain your body's finite resources (oxygen and blood, primarily) to the extent that runners turn to special, and at times peculiar, adaptation strategies. Runners training for the Badwater Ultramarathon will wheel their treadmill into the laundry room, disconnect the hose from the dryer vent, and then turn the appliance on at full blast during training runs in order to simulate the "hair dryer" effect, named for the notorious furnace-like winds runners face in Death Valley on July afternoons. More-daring runners will take it up a notch and layer on extra clothing, too. The ultra-paranoid will turn on a space heater in the room for good measure. Meanwhile, ultrarunners concerned with altitude deploy tents, masks, training camps, and all other methods of acclimation in order to prepare for racing through the thin air.

While all of these training interventions are well intended, few actually work. Some will have a net-zero effect, and several are actually counterproductive. So, before you go out and unhook your dryer in an attempt to battle Death Valley heat, let's paint some perspective on how heat and altitude actually affect performance and what you can do about them.

Fitness Is Your First Weapon

First things first: fitness is your primary weapon for optimal performance both at high altitude and in searing heat. The more fit you are, the more effective your natural cooling mechanisms are for hot environments and the more red blood cells you have to carry oxygen in high-altitude conditions. I mention this up front because it's important; none of the training interventions discussed below should be undertaken at the expense of building fitness. I mean it (so much so that I will repeat the point several times throughout this chapter). If your fancy altitude tent compromises your sleep and therefore your ability to complete (or recover from) hard workouts, skip sleeping in the tent. Similarly, running hard intervals in the middle of a hot day might provide a heat acclimation boost, but it will also limit that workout's effectiveness, so don't partake in that nonsense either. As you will see in Chapter 10, the total volume of workload you can do and the quality of the workouts you do are far more important than specific heat and altitude protocols. Yet, they are not mutually exclusive. So, if you can get the right amount of work in and perform either or both of these interventions, great! But, if you have to compromise workload or quality to do so, it's likely not a good trade-off.

With all of that said, let's dive into considerations and interventions for high-altitude and extreme-heat situations.

Altitude

Aside from heat (which I will talk about later in this chapter), altitude might be the scariest environmental factor for many trail and ultra-athletes. We host a running camp in May every year in Colorado Springs, Colorado, and

the most common thing we hear going into camp is, “I’m from sea level. I’m worried that I will be too slow at the camp because of the elevation.” While altitude is not everyone’s friend, it doesn’t have to be your foe either.

“There’s not enough air up here!” is something I myself have yelled while running in the mountains, and while it’s an incredibly popular phrase, it’s not entirely true. Despite a change in altitude from sea level to 8,000 feet, the composition of gases that make up the air around you remains the same: 20.93 percent oxygen, 0.03 percent carbon dioxide, and 79.04 percent nitrogen. So why does it feel so terrible up high? As you go up in elevation, there is a decrease in the partial pressure of oxygen, and your lungs rely on a pressure gradient for both inhalation and exhalation as well as the gas exchange between your alveoli (the small air sacs in your lungs) and capillary beds (Sinex and Chapman 2015). Basically, there isn’t “less air,” there’s less oxygen in a lungful of air due to changes in partial pressure at higher elevations.

PHYSIOLOGY AT A GLANCE

Despite these environmental challenges, our bodies are incredibly adaptable at the multisystemic and cellular levels to not only survive but thrive and perform at a high level. The research literature on human survival and performance is rich: from military interests to mountaineering feats, altitude has been inspiring physiologists for generations. This fascination with performance at high altitudes was heightened even more heading into the 1968 Summer Olympics in Mexico City (located at ~7,350 feet), where performing well at altitude meant the glory of Olympic medals for the countries and athletes competing.

So, what happens when you travel to higher altitudes? When you travel to altitude, your body needs to go through a series of adaptations very quickly to counter your decreased ability to transport oxygen efficiently. Remember that oxygen moves from the lungs to the bloodstream because of a pressure gradient, so the reduction in partial pressure means it’s harder to get oxygen into the bloodstream and then to both working muscle tissue and your brain. The adaptations you undergo are generally two-pronged: some take place in the first twenty-four hours at altitude, while others may take several weeks to fully develop. And while we may think we are limited by our lungs and respiratory system, the adaptations take place throughout our bodies and affect our cardiovascular and metabolic systems as well.

At a cardiovascular and hematologic level, changes can take place almost immediately. This is because your body is trying to artificially boost the density of your red blood cells through a process known as hemoconcentration. When this happens, your plasma volume can decrease by 10 to 25 percent, which in turn increases your hematocrit (the percentage of red blood cells—which contain hemoglobin, the oxygen-carrying component—per unit of blood). Ramping up hematocrit helps your body temporarily improve its ability to transport oxygen until you can produce more red blood cells to get the job done (Beidleman et al. 2017). This is also why your heart rate tends to be elevated for the first few days at higher elevation. As blood volume decreases, so does stroke volume (the amount of blood pumped out of the heart per beat), and your heart rate has to increase to keep cardiac output stable (cardiac output $[Q] = \text{HR} \times \text{stroke volume}$). This will normalize at rest and at lower aerobic intensities more quickly, but it is why you generally feel your worst from twenty-four to seventy-two hours after arriving at altitude. This is also why athletic programs recommend you keep your training intensity low (below 70 percent of $\text{VO}_2 \text{ max}$) when you first arrive at altitude to avoid increased fatigue or maladaptation (Sinex and Chapman 2015).

At a respiratory system level, the decrease in oxygen coming into your bloodstream from gas exchange in your lungs causes an increase in ventilation rate and depth of breathing. This increase in ventilation is important because of its downstream effects. When you breathe faster you exhale more carbon dioxide than normal. This in turn means there is less carbon dioxide in your blood and can eventually lead to respiratory alkalosis, or a pH imbalance (Sinex and Chapman 2015). Because our bodies like to operate within very finite parameters, this increase in blood pH causes our kidneys to get involved to correct-course and bring our body back to equilibrium. The kidneys do this primarily by increasing urine volume. The other things your kidneys do in response to hypoxia are produce and release more erythropoietin (EPO; Goldfarb-Rumyantzev and Alper 2014). EPO might sound familiar due to its illicit use as a performance-enhancing drug, but EPO is also naturally produced to help stimulate the body to make more red blood cells. The process of erythropoiesis (the production of red blood cells), stimulated by EPO, takes at least one week and as many as two to three weeks to allow red blood cells to mature

and become fully functional (Garvican-Lewis 2017). It is this long-term response that allows your blood plasma volume to not only return to normal but to also expand, and with this increase in red blood cells there are also increases in your hematocrit and hemoglobin and ultimately the desired adaptation of carrying more oxygen. This entire cascade of events is described in Figure 7.1.

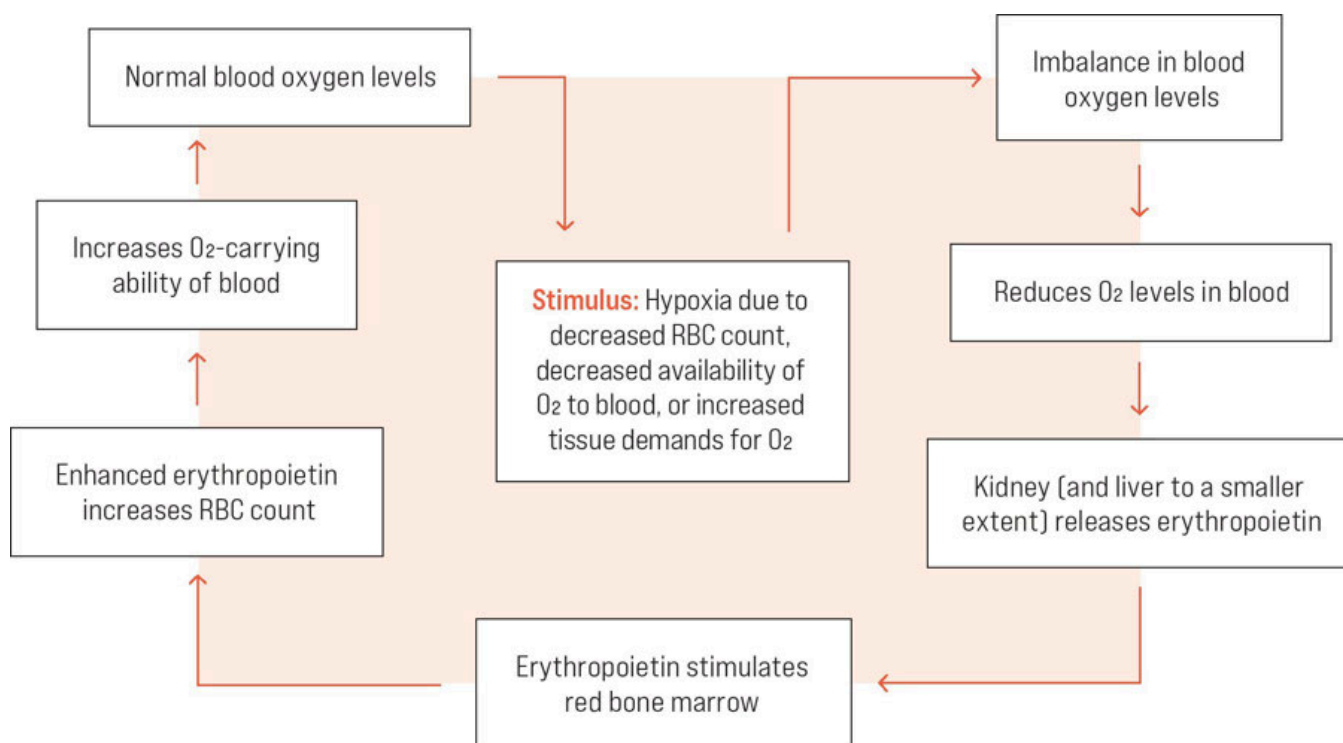


FIGURE 7.1 Hematologic effects to altitude exposure.

ACCLIMATION PROTOCOL FOR ALTITUDE

When it comes to methods for altitude acclimation, there are two primary contenders: “live high, train high” and “live high, train low.” If you have the ability, time, and means to either put together your own high-altitude training camp, or if you live at moderate to high altitude, you might be interested in understanding what the ideal setup is. The general consensus in the research community is that the “live high, train low” method will lead to the greatest hematologic and performance benefits (Constantini, Wilhite, and Chapman 2017). This is because you get both the long-term altitude exposure throughout the day and overnight and the higher-quality workouts performed at lower elevations. Of course, if you are really lucky, you fall into the “live high, train high/ train low” camp, in which you live high, do low-intensity workouts up high, and then do high-intensity workouts down low, allowing you to perform high-quality workouts while utilizing maximum recovery. However, there are fairly few training sites that allow this type of training to be done conveniently (living high with a short commute to running low). These sites include Park City, Utah; Flagstaff, Arizona; Mammoth Lakes, California; Big Bear Lake, California; Lake Tahoe, Nevada/California; Cloudcroft, New Mexico; and Sierra Nevada, Spain (Constantini, Wilhite, and Chapman 2017). They might sound familiar, as many professional running groups use them as training hubs, fulltime residences, or high-altitude camps.

What does all of this mean for you as you get ready for the Leadville 100 while living in the Midwest? First, don’t worry, you are not alone. Second, there are few things to consider. When we think about altitude acclimation, there are two primary methods: using an altitude tent or finding a natural high-altitude environment. No matter which one you use (or have access to), effectiveness often comes down to dosage: how high for how long.

If you can use a natural environment, it is important to have your chronic exposure take place above 5,900 ft (1,800 m) and below 9,840 ft (3,000 m). It's important to point out that there isn't anything inherently negative about being above 9,840 ft; there just do not appear to be any added benefits, hematologically speaking. Additionally, with an increase in elevation, there comes an increased risk of acute mountain sickness, so think of the altitude cap as balancing optimal adaptations with relative safety.

	LIVE HIGH, TRAIN HIGH	LIVE HIGH, TRAIN LOW	LIVE HIGH, TRAIN HIGH / LOW
Living	Living between 6,840–8,050 ft	Living between 6,840–8,050 ft	Living between 6,840–8,050 ft
Training	Training at or above 6,000 ft	Training below 4,100 ft	Train at low intensities between 6,840–8,050 ft, and high intensity workouts below 4,100 ft*
Duration of Protocol	~21–28 days	~21–28 days	~21–28 days

TABLE 7.1 Different altitude training protocols.

*Living between 6,840–8,050 ft is optimal for hematological responses while not impairing recovery and sleep. Moving high-intensity runs below 4,100 ft is also optimal because it allows you to preserve the ability to effectively deliver oxygen to working muscles, maintaining the quality of higher-intensity sessions.

As far as dose is concerned, the general recommendation is a minimum of two weeks, and ideally up to four weeks, to reap maximum performance benefits (Constantini, Wilhite, and Chapman 2017). This might not be realistic when you are looking at travel plans for your upcoming A race, and that's where the one-week minimum recommendation comes into play. This recommendation is not based on maximizing performance benefits but rather avoiding feeling terrible on race day. As I mentioned earlier, athletes generally feel worst during the twenty-four- to seventy-two-hour period after arriving at altitude, so getting over that hump will at least spare you the added hindrance of your body trying to make its initial adaptations.

Finally, it's important to note that traveling to altitude above 8,202 ft (2,500 m) does not come without general health risks such as developing acute high-altitude illness, which covers several syndromes such as acute mountain sickness (which usually resolves without treatment) or, more seriously (potentially fatally), high-altitude pulmonary edema (swelling of the airway) and high-altitude cerebral edema (swelling of the brain). Because of this, it's important to consult with your primary care provider if you have any questions or concerns before traveling to altitude.

When it comes to using an altitude tent, the time course for adaptation remains fairly constant, with three to four weeks of exposure needed to produce any hematological changes. However, the biggest concern around this strategy is that you're not actually spending a full three to four weeks in the altitude tent; at most you are probably spending eight to ten hours a day exposed to hypoxia. Researchers are still struggling to discern meaningful hematological changes in athletes utilizing altitude tents while living at altitudes lower than 5,900 ft. That said, if you live at the cusp of the minimum elevation required for chronic altitude exposure (~5,500–6,500 ft), several studies have shown that utilizing an altitude tent for nine hours a night at a simulated altitude (8,200–9,840 ft) was enough exposure to produce positive adaptations (Carr et al. 2015; Garvican-Lewis et al. 2015).

What does that mean for altitude tents? I think the verdict is still very much out. But as a coach I weigh the following questions before recommending them on an individual basis: Can the athlete afford the tent? Does it decrease their training quality by impacting their quality of sleep? At what elevation do they currently live? And, can they make it out to their race at least one week prior to the gun going off?

IS AN ALTITUDE INTERVENTION A GOOD IDEA?

There is one more important piece of this puzzle that is worth mentioning. Like everything else in physiology, there is some degree of individual variation when it comes to altitude acclimation. Over the past twenty years there have been terms tossed around like “responders vs nonresponders,” and while it does appear that people are affected differently by chronic exposure to altitude, oftentimes it’s because we are missing an underlying cause. These underlying causes can inhibit an individual’s ability to produce a normal response to altitude exposure, including the expected increases in EPO and hemoglobin. This does not mean an individual may never be a responder to altitude, but it does reinforce the importance of context when looking at research outcomes.

Factors we know can easily inhibit your ability to adapt to altitude include injury, illness, and iron status. Injury and illness can both create inflammation in the body that naturally reduces your hemoglobin at sea level, before you even leave for altitude. Additionally, injury and illness can elevate cortisol levels, which blunts adaptation. This is largely why physiologists recommend avoiding high-intensity training (which also raises cortisol levels) in the days before traveling to altitude. Finally, low iron status, specifically stored iron measured as serum ferritin, can impact your body’s ability to create hematological adaptations at altitude. In studies evaluating runners at high-altitude camps, researchers found that athletes whose serum ferritin was less than 35 ng/mL could not produce an increase in red blood cell mass even after four weeks of altitude exposure (Constantini, Wilhite, and Chapman 2017).

PRE-RACE ALTITUDE STRATEGIES

This is where I remind you that the absolute best thing you can do, no matter the elevation of your hometown, is to make sure you are as fit as possible before going to a high-altitude race. No amount of magic, sleeping in a tent, prayer, or time at altitude is going to make up for a lack of fitness and other race-specific factors. Along with great fitness, here are three things you can do to optimize your race-day performance:

GET AN IRON PANEL THREE MONTHS OUT: Three months before going to elevation, meet with your primary care provider and get an iron panel done. (This should be done before taking iron supplements, because consuming too much iron can be toxic.) Three months gives you time to benefit from increased iron intake (if necessary) so you have the tools to adapt to altitude for a race, training camp, or time spent in an altitude tent.

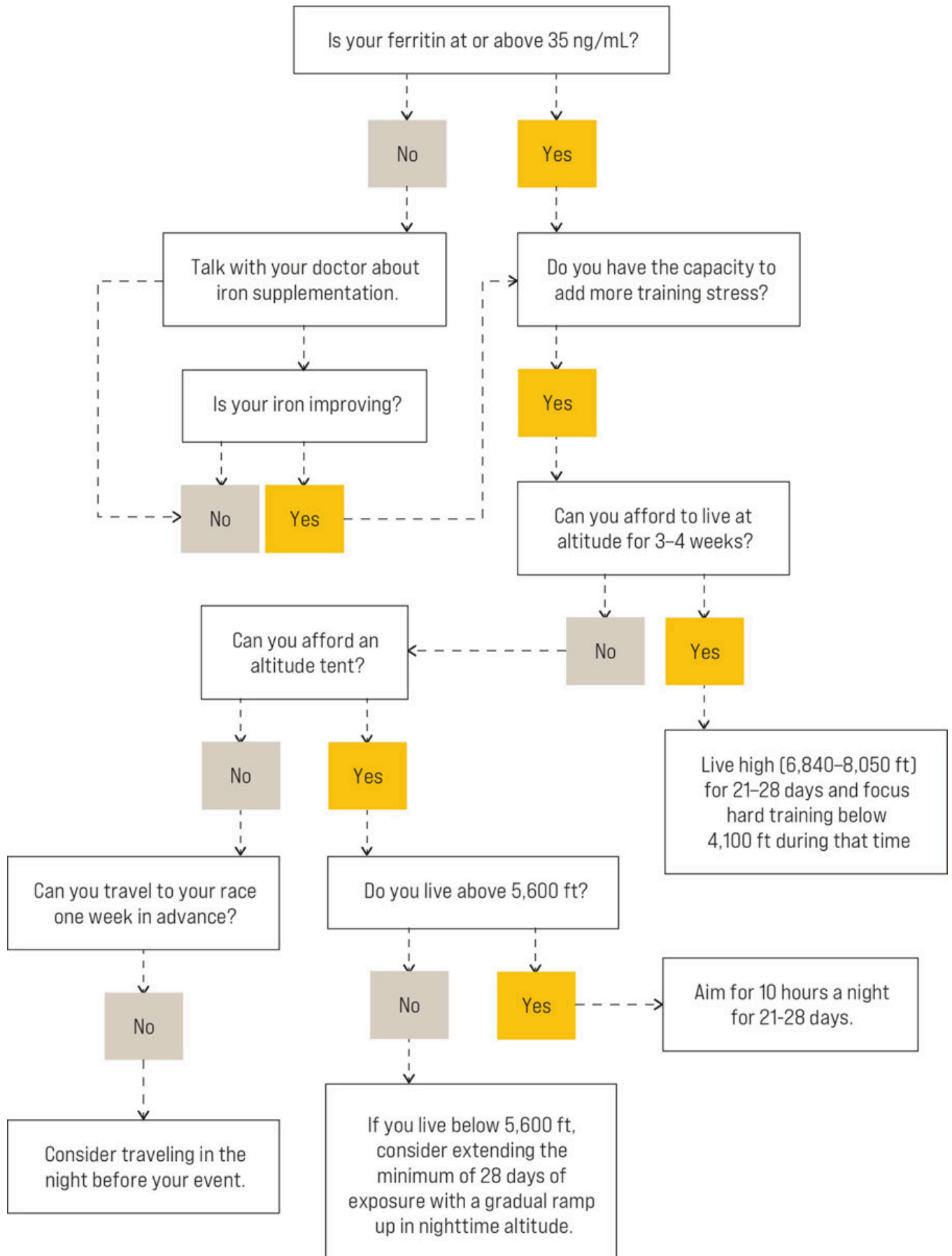


FIGURE 7.2 Are you a good candidate for altitude training?

TRY TO ARRIVE AT YOUR RACE EARLY: If you can't get there early, arriving the night before or morning of the event might allow you to fit your race into that twenty-four-hour window before your body ramps up the responses to altitude that make people feel miserable. General recommendations are to arrive three to five days early for races at 1,640–6,561 ft, one to two weeks for races at 6,561–9,842 ft, and at least three weeks for races over 9,842 ft (Chapman, Stickford, and Levine 2010).

ORGANIZE A TRIAL RUN: If possible, go to altitude for a camp ahead of a high-altitude A race. The benefits are mostly experiential, giving you an understanding of how your body responds to running at altitude. This is also a good time to figure out the adjustments you'll need to make to both fueling strategies and pacing strategies in order to (hopefully) avoid any unpleasant surprises come race day.

Heat

Many of you have probably thought or heard someone say, "I'm bad in the heat." I've read it in training logs and seen it in the eyes of runners coming into Michigan Bluff during the Western States 100. But here's the thing: although you might not love the heat, your body is clever and can definitely get better, both physiologically and psychologically, at running in hot conditions. Additionally, there are strategies you can employ before and during your event to make sure you don't melt, crumble, or end up in the medical tent. As with altitude, the heat doesn't care about how tough you are. How well you prepare for heat can make the difference when it comes to finding the finish line.

PHYSIOLOGY AT A GLANCE

As humans we are inherently inefficient beings. The truth is, only a fraction of the work you do ends up propelling you down the trail; the majority of it simply generates heat that doesn't really have anywhere to go. As a result, we rely first and foremost on our body's internal capacity to maintain our ideal core body temperature (37°C/98.6°F), and then we can then supplement that capacity with actions like putting on more or less clothes, etc. When your body temperature drops, your blood vessels constrict, keeping warmth near your core to protect your vital organs, and you begin to shiver in order to produce heat. In contrast, when your body temperature rises, your blood vessels dilate, sending more blood to the surface of the skin to dissipate heat into the environment and activating your sweat glands to provide water for evaporative cooling.

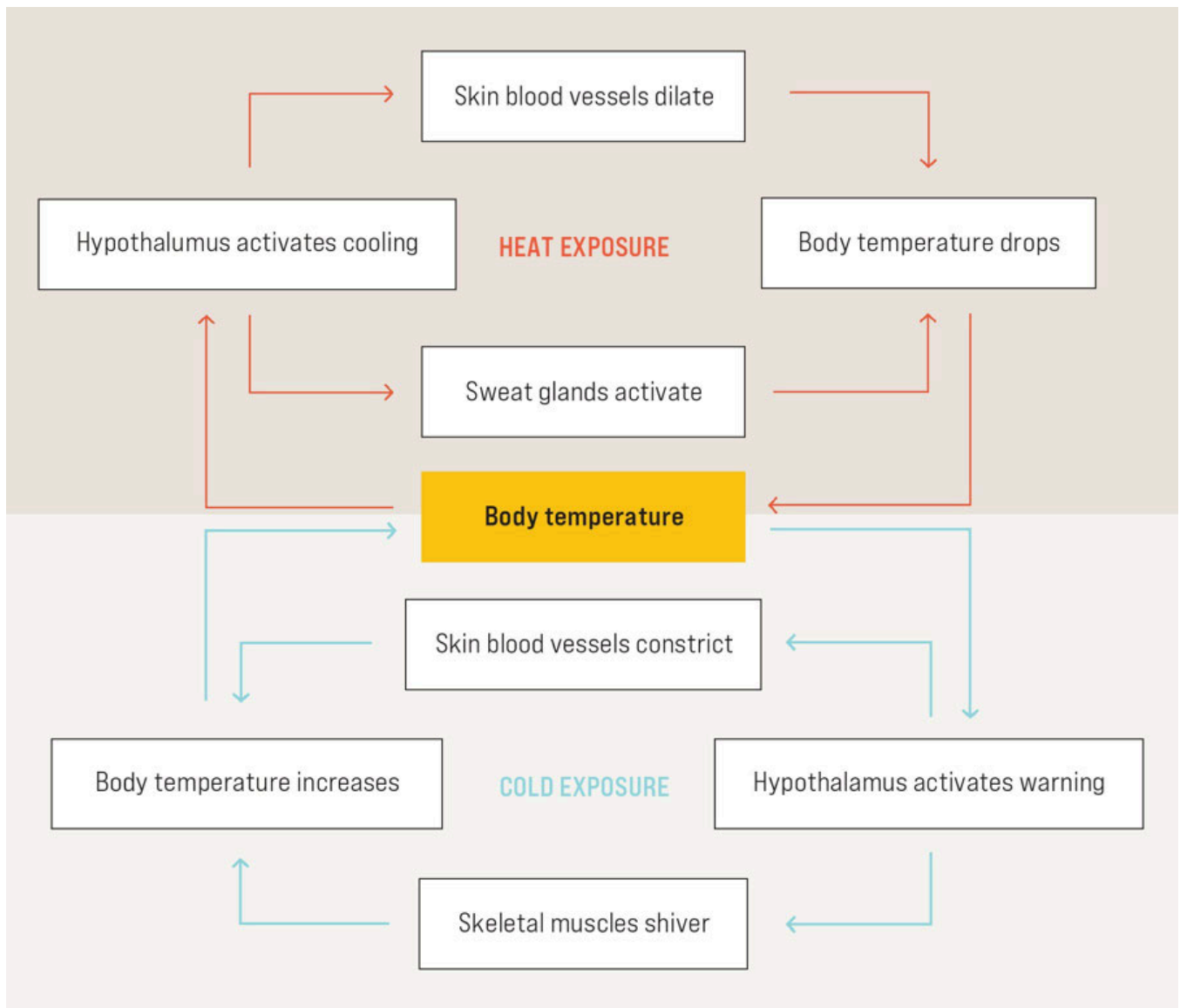


FIGURE 7.3 Physiological response to temperature change.

Again, there are two major factors at play when exercising and racing in the heat: the heat you create and the heat you're exposed to. Like I mentioned before, your body produces a lot of heat doing metabolic work when you run—as much as 5 to 15 times more than at rest—and it will try to dissipate as much of that as possible to maintain a relatively stable core temperature (Sawka et al. 1993). Additionally, you are running in a hot environment, which adds more heat strain to your metabolic mess.

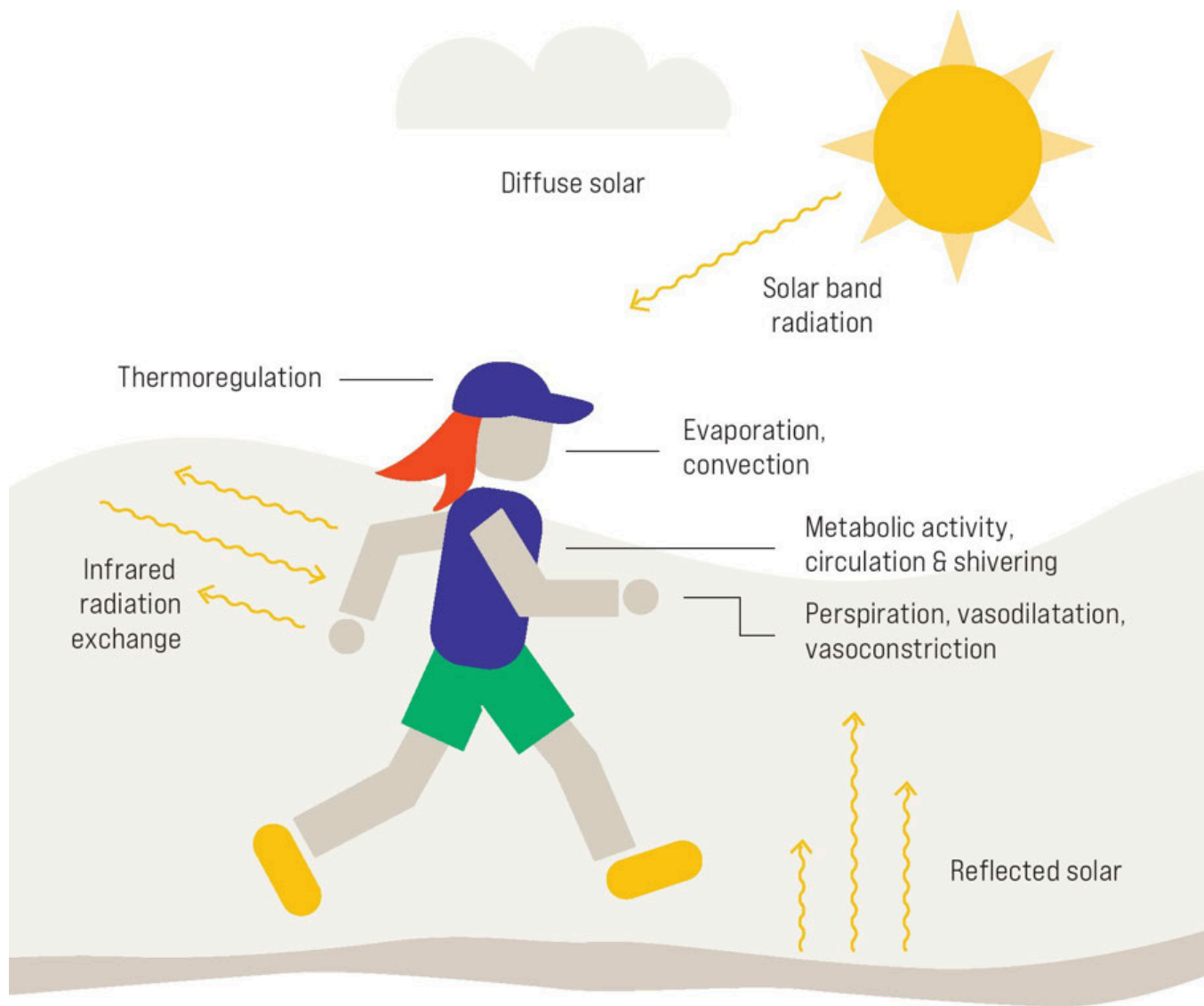


FIGURE 7.4 Thermoregulation of the human body.

Where you are running matters too. Some examples include an environment where ambient air temperature is greatly influenced by the surface you are running on (like at the Badwater 135 where the pavement reflects the heat of the sun), you're in direct vs indirect sunlight, or you're exposed to wind from varying directions. The environment impacts the four categories of heat transfer: conduction (direct transfer of energy), convection (air or water moving over your skin), radiation (direct or indirect transfer of heat from electromagnetic waves), and evaporation (conversion of liquid to gas; Sawka et al. 2011). These methods of transfer work together to either heat you up or cool you off, but their effect is largely dependent on the temperature gradient between you and the surrounding environment, because heat moves along a gradient (high to low).

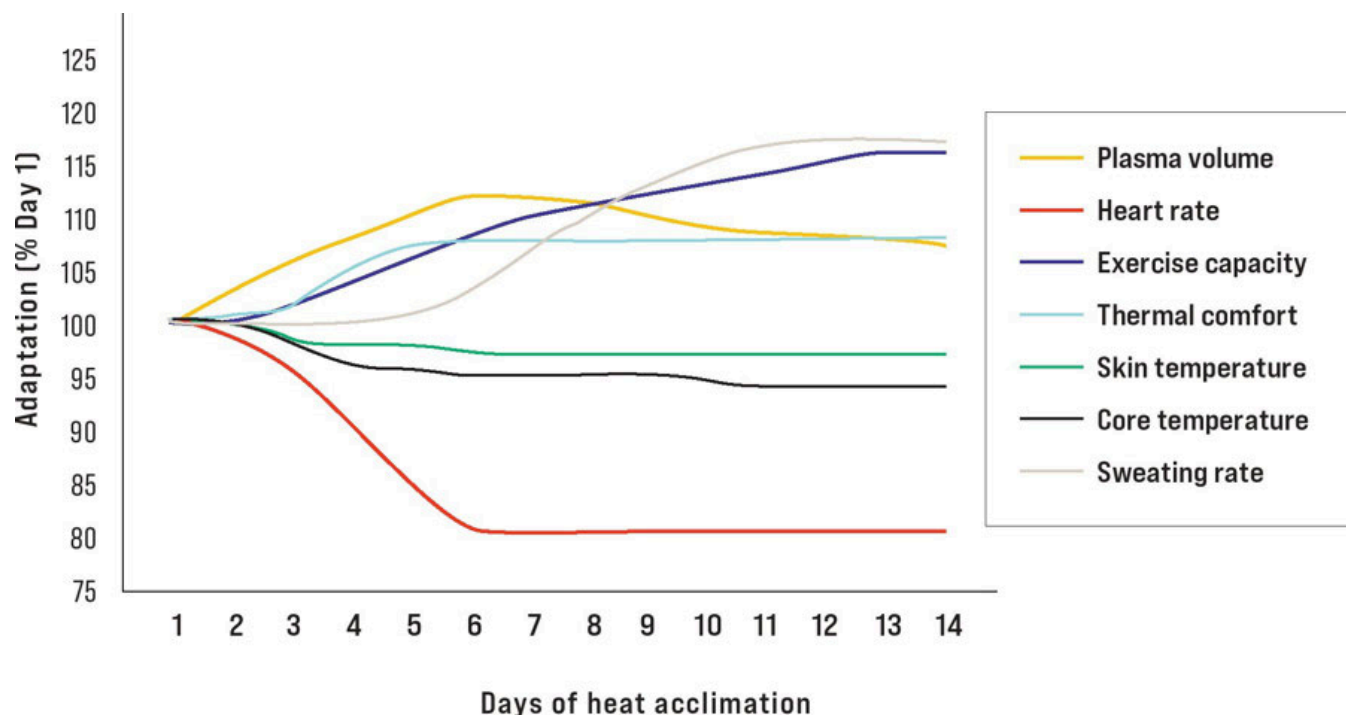


FIGURE 7.5 Time course of adaptations from heat acclimation. Adapted from Périard, Racinais, and Sawka 2015.

Acclimation, the process of your body getting more efficient and adjusting to your environment over time, applies to both altitude and heat. It is our clever capability to maintain, survive, and thrive across a wide range of environmental conditions. Heat acclimation leads to internal changes that enable the body to function better in hot environments. The key changes that your body makes after chronic, daily heat exposure are an earlier onset of the sweat response, an increase in sweat rate and a corresponding decrease in sweat sodium concentration, better reabsorption of sodium and chloride through the sweat duct, lower skin and core temperature thanks to blood plasma volume expansion, lower heart rate during exercise (or, rather, a return to normal heart rates), increased fluid and cardiovascular stability, and a decrease in the metabolic cost of exercise (Périard, Racinais, and Sawka 2015; Rivas et al. 2017). Combine all these factors and you have the most important adaptation of all: the ability to tolerate and feel more comfortable in the heat come race day.

ACCLIMATION METHODS

You might be looking to heat acclimation to perform better at a hot race or to jumpstart your summer training as the temperature creeps up, so what methods are available to you? As a coach I ask myself, “How do I elicit the adaptation I want with the lowest dose or minimum intervention possible?” This is important, because anything we add to your training is stress, and when adding stress there are always trade-offs. What do we take away to make room for this additional stress? We know from the literature that all of the following strategies can work, but the one that is best for you has a lot to do with what you have at your disposal and what you can tolerate.

What we know is that if you repeatedly, day after day, increase your core body temperature through exercise, environmental stress, passive interventions (sauna or hot-water immersion), or all of the above, you induce physiological adaptations that will increase your thermal tolerance (Minett et al. 2016).

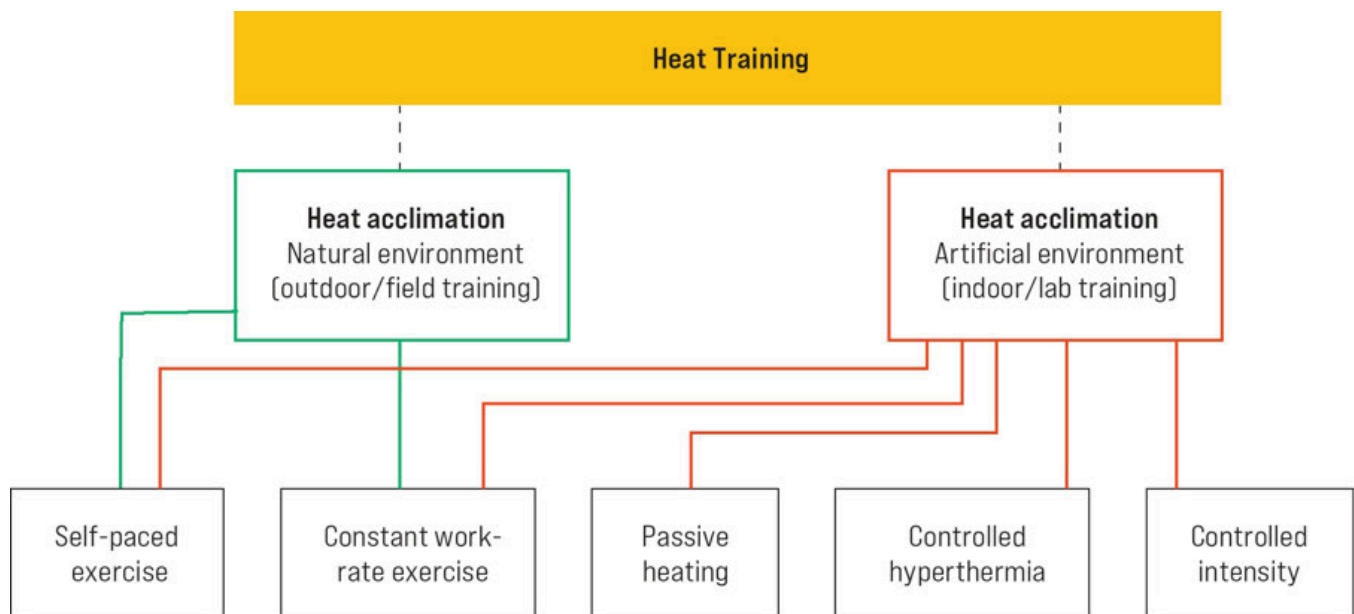
The methods most commonly used by athletes are natural heat acclimation (exercising in hot ambient conditions), controlled hyperthermia (overdressing), controlled intensity (use of exercise in a climatic chamber), and passive heating (postexercise heating via sauna or hot-water immersion; Daanen, Racinais, and Périard 2017). All of these

methods work but take varying amounts of time to increase your core body temperature to ~101°F (38.5°C), where it should stay for approximately ninety minutes. Additionally, some of these methods are time-consuming and inherently uncomfortable and could even be dangerous.

There have been numerous studies dating back to the 1960s that look at sessions ranging from 50 to 100 min of daily heat exposure, twice-daily heat exposure, trials lasting anywhere from five to twenty-seven days, and combinations of the aforementioned methods. The summary of this research is that the end results were more or less the same: they all work, and most importantly, more is not better.

PRIORITY OF METHODS

There is a hierarchy of heat acclimation protocols, and it's a little bit like "Goldilocks and the Three Bears": The one that fits just right is the one that allows you to maintain the highest quality training. Like everything in a performance physiology setting, we are looking to achieve the desired adaptations (boost performance) with the smallest dose possible.



METHOD	ACTIVITY EXAMPLE	WORK RATE	CONDITIONS	DURATION
Self-paced exercise	Football, or running	Variable, or self-selected	Variable outdoor, or indoor 40°C/104°F, 40% relative humidity	60–90 min
Constant work-rate exercise	Cycling ergometer, or marching	60% VO ₂ max (185 W) or 6 km/h	Indoor 40°C/104°F, 40% relative humidity, or variable outdoor	60–90 min
Passive heating	Water immersion, or sauna bathing	N/A	Water 40–42°C/ 104–107.6°F or sauna 70–90°C/158–194°F	45–60 min, or intermittent for 30 min
Controlled hyperthermia	Resting and/or exercising to maintain core temperature at 38.5°C	Variable	40°C/104°F, 40% relative humidity	60–90 min
Controlled intensity	Cycle ergometer	65% VO ₂ max heart rate (145 bpm)	40°C/104°F, 40% relative humidity	60–90 min

FIGURE 7.6 Different heat acclimation methods. Adapted from Daanen, Racinais, and Périard 2017. Self-paced exercise



FIGURE 7.7 Hierarchy of heat acclimation protocols.

Active heating by overdressing or by exercising in a controlled environment like a climatic chamber is effective, but are you sacrificing workout quality by cutting the volume or intensity? Likely both. You can also spend a long time (sixty to ninety minutes broken up as needed) in a dry sauna, wet sauna (a steam room), or hot tub to get the desired effects of heat acclimation, but that's a lot of additional time and stress added to your daily training regimen. When we take these two factors—dose and time needed—into consideration, passive heating (dry sauna, wet sauna, or hot tub) immediately postexercise rises to the top of the pile. This strategy allows you to maintain the quality of your workouts in the natural environment and spend less time in the actual heat because you are entering “pre-warmed” by the exercise.

It's important to note that inducing hyperthermia has inherent risks, and it is important to monitor yourself throughout any protocol. This can include monitoring core body temperature and hydration status (either by weighing yourself pre- and post-sauna to account for water loss or by using urinalysis strips to ensure you rehydrate adequately post-sauna). Additionally, you should consult your primary care provider if you have any questions or concerns.

ACCLIMATION PROTOCOLS

Passive heating best combines the temperature threshold needed for adaptation to take place with a small and manageable additional stressor on top of training workload. The following protocol illustrates how we help athletes as they prep for hot races like Western States 100, Transgrancanaria, and The Coastal Challenge.

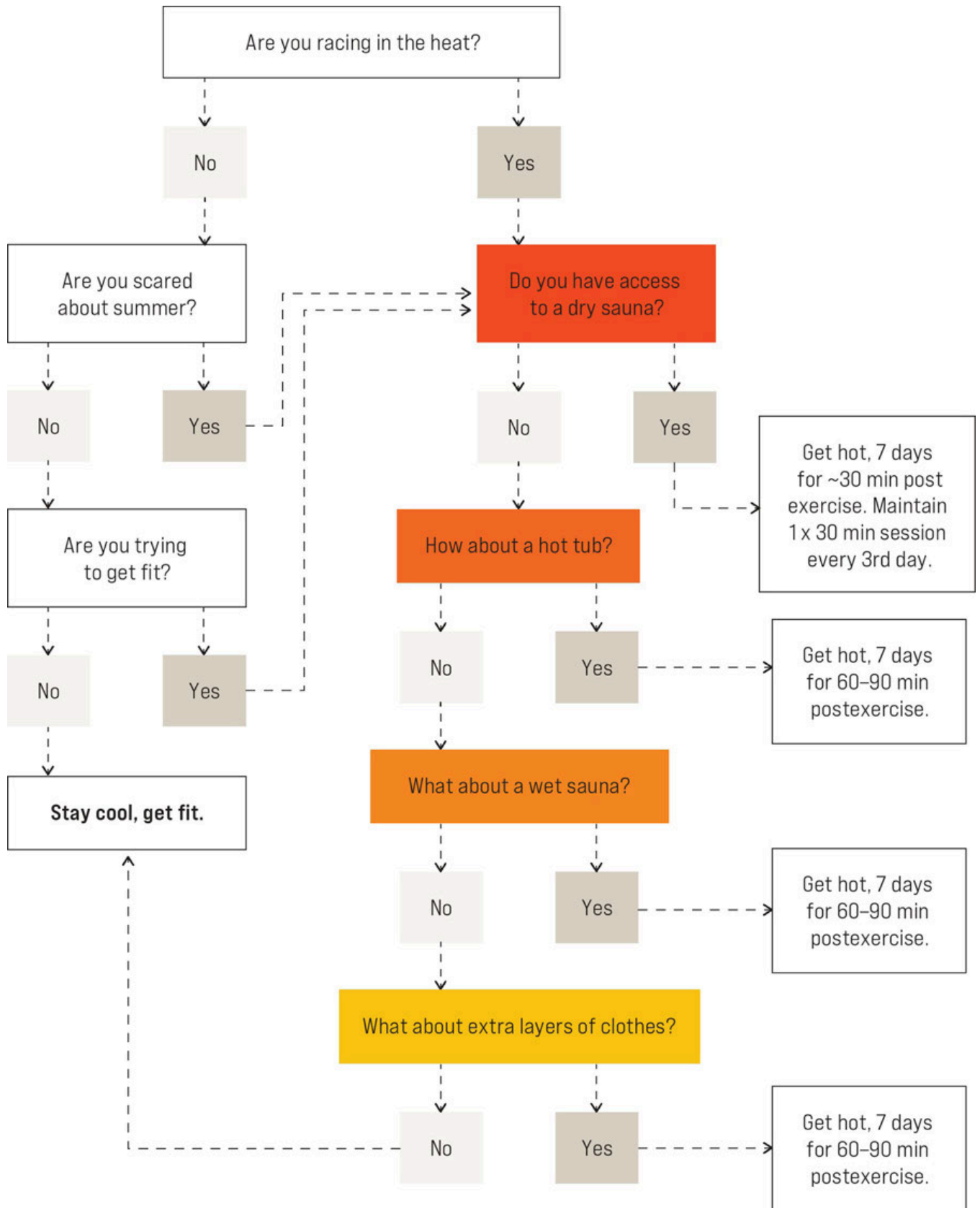


FIGURE 7.8 How to choose the right heat acclimation protocol.

MON	TUES	WEDS	THURS	FRI	SAT	SUN
5	6	7	8	9	10	11
Sauna after exercise for 15 min	Sauna after exercise for 15 min	Sauna after exercise for 15-30 min	Sauna after exercise for 15-30 min	Sauna after exercise for 30 min	Sauna after exercise for 30 min	Sauna after exercise for 30 min
12	13	14	15	16	17	18
Sauna after exercise for 30 min	Sauna after exercise for 30 min	Sauna after exercise for 30 min			Sauna after exercise for 15-30 min	
19	20	21	22	23	24	25
Sauna after exercise for 15-30 min					Race Day	

FIGURE 7.9 An example of a heat acclimation protocol using a sauna.

Things to keep in mind:

- Heat acclimation will make you tired. Our aim is to minimize this consequence.
- You need seven to ten days in a row, ideally, to elicit positive adaptations.
- After heat acclimation is achieved, you can maintain these adaptations by adding exposure to heat again once every three days.
- The average degradation of heat acclimation is ~2.5 percent a day after the first forty-eight hours without heat exposure. If you were to do a two-week taper with no heat exposure, you could expect a decay in heart rate control of ~35 percent, a decay in core temperature control of ~6 percent, and a decay in sweat rate improvements of ~30 percent. If you are traveling to an event in a hot environment, it is important to know you can generally reacclimate in as little as four days.
- A heat acclimation schedule can be adjusted to align with the “most manageable” training (i.e., it’s not ideal to do this on top of peak volume training).
- In the following examples, I use “sauna” as the preferred method of choice. This is because of its efficacy, utility, and accessibility. A hot-water immersion bath would be a close second choice.

	MON	TUES	WEDS	THURS	FRI	SAT	SUN
6 weeks before race	5	6	7	8	9	10	11
	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min		Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	
5 weeks before race	12	13	14	15	16	17	18
	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min		Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min
2-4 weeks before race	2 sauna sessions per week						
	9	10	11	12	13	14	15
	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min		Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min	Sauna after exercise for 20-30 min
	9	10	11	12	13	14	15
	Sauna after exercise for 20-30 min	Sauna after exercise for 15-20 min		Sauna after exercise for 15-20 min		Race Day	

FIGURE 7.10 Two-phase sauna acclimation protocol.

Protocol #1: Single Exposure Leading up to the Race

If you have limited time and resources to gain access to a sauna, adding in a heat acclimation protocol in the several days leading up to an event is going to be your best choice. However, take extra precaution that you taper more than you normally would, as the sauna will add additional stress on your system, which is contraindicated for the taper.

Protocol #2: Repeated Exposure

If you have more regular access to a sauna and if your training leading up to an event is going particularly well, this protocol might be more effective for you. This protocol, while more complicated, mitigates the risk of overdoing the sauna exposure close to the race. For this protocol, you take full advantage of a longer period of time by acclimating in two separate bouts. One occurs approximately six weeks out from the event, and the second occurs in the final week of the event. In between, you will have several heat “maintenance sessions” approximately every third or fourth day so that when you are ready for the final re-acclimation period it can be faster and more effective. These maintenance sessions also partially mitigate the ~2.5 percent per day decay in

heat acclimation mentioned previously. Additionally, we know that repeated exposure to hot environments have an additive effect. So, you might get a slightly better adaptation from this protocol. The downside of this protocol is that the first acclimation period (about six weeks out from an event) can coincide with heavy and important training. If this is the case for you, consider protocol #1 so you are not adding so much stress all at the same time.

Race-Day Considerations

So, you've done the work, you're fit, and you've spent the time to be as ready as possible to handle the heat. Now what about when you're actually racing? Even after effective heat acclimation, there are still strategies you can employ to both improve performance and avoid a medical emergency like heat stroke. We refer to these strategies as "during event cooling."

During-event cooling strategies all aim to rapidly remove excess heat from the body in hopes of creating more heat-storage capacity (Bongers, Hopman, and Eijssvogels 2017). The goal is not so much to actually lower your core body temperature, but rather to keep your skin temperature low enough to maintain a temperature gradient between your core and skin. This encourages blood to be sent to your skin (via vasodilation) to release heat into the environment. Maintaining this delicate balance is like walking a tightrope.

Strategies you can employ in your next race include packing ice on your torso, icing your neck and head (ice bandanas, collars, and hats), icing your peripheral arteries (ice down your arm sleeves), wearing loose and light-colored clothing to facilitate heat dissipation, limiting conductive heat (wearing light colors), increasing evaporation (dump water on yourself), and drinking cool fluids (Tyler, Sunderland, and Cheung 2013). Not to be outshone by more exciting strategies, I cannot stress enough how important it is to maintain proper hydration status—it is of utmost importance to keep your sweat response intact so you can utilize evaporative cooling and keep your core temperature below critical levels. Finally, a really effective way to cool down is to stop running (because remember, you are actively producing heat when you run). If you need to, you can save your race by walking slowly or stopping for ten, twenty, or thirty minutes to ingest fluids and food while you get some ice or cold water on your body.

KEY POINTS ON ADAPTING TO THE ENVIRONMENT

Ultramarathons are frequently contested in hot and high-altitude environments. Heat and altitude are both worth taking into consideration for training.

Fitness is your first weapon in combatting heat and altitude.

Your body will adapt to altitude through a host of activities, primarily by producing more red blood cells and expanding plasma volume.

When initially arriving at altitude, your body will respond with an elevated heart rate, which normalizes after twenty-four to seventy-two hours.

There are three methods of acclimating to altitude:

- **Live High, Train High**
- **Live High, Train Low**
- **Live High, Train High/Train Low (low-intensity sessions at high altitude, high-intensity sessions at low altitude)**
- **Each of these protocols can last from twenty-one to twenty-eight days.**
- **The most efficacious is the Live High, Train High/Train Low protocol.**

Not all athletes are good candidates for altitude camps. Athletes with low ferritin levels (< 35 ng/ml) have room to improve through normal training, and athletes who cannot logistically orchestrate an altitude camp should focus on improving through other means.

No matter what elevation you are training at, if you are going to a race at altitude you should conduct an iron panel three months out, try to arrive at the race early (how early depends on the altitude of the race), and organize a trial run at the altitude of your race to see how you react. (Keep in mind there are real risks associated with the development of acute high-altitude illnesses at elevations above 2,500 m/8,202 ft).

You produce five to fifteen times more heat when you run vs at rest. Your body will dissipate as much of this heat as possible through conduction, convection, radiation, and evaporation.

Commonly used methods to acclimate to the heat are natural heat acclimation, overdressing, exercising in a climatic chamber, and passive heating via a sauna or hotwater bath. (Keep in mind it's important to monitor core temperature and hydration status during any protocol.)

All of these methods work, but passive heating (sauna or hot-water immersion) will be the most practical and efficacious and will interfere with training the least.

Even if you are heat acclimated, athletes competing in hot environments should employ race-day cooling methods such as packing ice on your neck/torso, dousing in cool water, and consuming cool drinks.



